INCH-POUND
A-A-52106B
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SUPERSEDING
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November 19, 2019

COMMERCIAL ITEM DESCRIPTION

CLOTH, TWILL OR PLAIN WEAVE, POLYESTER AND POLYESTER BLEND

The General Services Administration has authorized the use of the Commercial Item Description (CID) by all federal agencies.

1. SCOPE. This CID covers the requirements for twill or plain weave polyester and polyester blend cloth.

2. CLASSIFICATION. This CID covers the following classes as specified.

2.1 Classes.

Class 1 - Heavy weight, 4.3 ounces per square yard
Class 2 - Medium weight, 2.9 ounces per square yard
Class 3 - Light weight, 2.2 ounces per square yard
Class 4 - Light weight, 2.4 ounces per square yard
Class 5 - Medium weight, 3.0 ounces per square yard
Class 6 - Medium to heavy weight, 3.6 ounce per square yard

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FSC 8305

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3. SALIENT CHARACTERISTICS.

3.1 General description. The Class 1 and 2 cloths shall be a two (2) up-one (1) down twill or a plain weave using stable or textured polyester yarns or using yarns spun from staple polyester and carded cotton or stable rayon. The Class 3 cloth shall be a two (2) up-one (1) down right hand twill weave using filament polyester yarns. The Class 4 cloth shall be a two (2) up-one (1) down right hand twill using 100 percent filament polyester single ply yarns. The Class 5 cloth shall be a five (5) up-one (1) down herringbone twill using 100 percent filament polyester single ply yarns. The Class 6 cloth shall be a rep weave using filament polyester yarns. For all other shuttleless looms, the width measurement shall be made between the last warp yarns on each side excluding the protruding fringe(s).

The width of the cloth shall be as specified (see 7.5) and shall be the minimum acceptable width inclusive of the selvage when fly-shuttle or shuttleless looms with tuck-in acceptable width inclusive of the selvage when fly-shuttle or shuttleless looms with tuck-in selvage are used.

3.1.1 Color. The color of the finished cloth shall be natural, bleached or dyed as specified in the contract or purchase order (see 7.5). The finished cloth shall conform to the colorfastness requirements specified in Table I when tested in accordance with test methods in Table I.

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TABLE I. Physical characteristics and test methods.

Characteristic	Requirement						Test method
	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	
Fiber content (percent):							
Polyester	47 - 100	47 - 100	100	100	100	100	AATCC TM20A
Cotton	0 - 53	0 - 53	0	0	0	0	AATCC TM20A
Rayon	0 - 53	0 - 53	0	0	0	0	AATCC TM20A (see 7.2.1)
Colorfastness (min.):							
Perspiration	3-4	3-4	3-4	3-4	3-4	3-4	AATCC TM15 1/
Dry Heat	3-4	3-4	3-4	3-4	3-4	3-4	AATCC TM117 2/, 3/
Wet drycleaning	3-4	3-4	3-4	3-4	3-4	3-4	AATCC TM132 3/
Crocking (wet and dry):							
(light colors)	3.5	3.5	3.5	3.5	3.5	3.5	AATCC TM8 4/
(black or dark colors)	1.0	1.0	1.0	1.0	1.0	3.0	
Weight, oz./sq.yd. (min.)	4.3 5/	2.9	2.2	2.4	3.0	3.6	ASTM D3776/D3776M
Yarns per inch (min.)							
Warp	73 5/	73	120	124	130	140	ASTM D3775
Filling	46 5/	46	66	76	92	100	
Breaking strength, lbs. (min.)							
Warp	72	58	100	140	160	100	ASTM D5034
Filling	55	44	75	110	130	75	
Dimensional stability (percent) (max.)							
Warp	3.0	3.0	3.0	3.0	3.0	3.0	AATCC TM158
Filling	3.0	3.0	3.0	3.0	3.0	3.0	
Tearing Strength, lbs. (min.)							
Warp	---	---	---	5.0	14.0	---	ASTM D2261
Filling	---	---	---	3.0	10.5	---	
Seam efficiency (percentage) (min.)	70	70	---	---	---	---	ASTM D1683 6/
Toxicity	See 3.8	See 3.8	See 3.8	See 3.8	See 3.8	See 3.8	See 3.8.1

1/ Rated using the AATCC EP1 and AATCC EP2.
2/ The test shall be performed at 325 (± 6) ºF.
3/ Rated using the AATCC EP1.

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4/ Rated using the AATCC EP8.
5/ For 100 percent polyester cloth a 15 percent minus tolerance is permitted.
6/ The needle shall measure 0.044 inch (± 0.001) inch across the blade at the eye. The thread shall be polyester core; cotton, or polyester-covered in accordance with A-A-50199,
Tex size 36-45 for the needle and Tex size 31-35 for the looper or spun polyester in accordance with A-A-59989, Tex size 30; 2 or 3 ply for the needle and Tex size 40; 2 or 3 ply for the looper. (Not applicable for 100 percent polyester cloth).

3.2 Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 7.4.1) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the finished cloth shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65 (see 7.4.2). The finished cloth shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

3.3 Physical characteristics and requirements. The finished cloth shall conform to the physical and colorfastness characteristics specified in Table I when tested in accordance with methods in Table I.

3.4 Labile sulfur (for blends only with cotton). The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The fabric shall be “Free” of labile sulfur when tested as specified in 3.4.1.

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3.4.1 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail”.

3.5 pH. The pH of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested in accordance with AATCC TM81.

3.6 Length and put-up. For Government procurement only, unless otherwise specified (see 7.7), the finished cloth shall be furnished in continuous lengths of not less than 40 yards and shall be put-up full width on rolls as specified in contract or procurement documents.

3.7 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Title 16 CFR Part 303, Rules and Regulations under the Textile Fiber Products Identification Act, (see 7.1.2).

3.8 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact (see 3.8.1).

3.8.1 Toxicity testing. When the toxicity requirement (see 3.8) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required, (see 7.1.5 and 7.5.)

3.9 Workmanship. The finished cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the contractor’s own quality assurance standards and the quality assurance standards defined by the technical data in the bid package.

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4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation FAR, or the sustainable acquisition of EO 13693; as applicable (see 7.1.3 and 7.1.4).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The finished cloth provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. This finished cloth must have been sold to the government or commercial market.

5.3 Inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (see 7.2.2).

5.4 Visual examination. The finished cloth shall be examined for the defects listed in Table II as defined in ASTM D3990:

TABLE II. Visual examination defects.

Material defects: Any spot, or stain more than 1-inch in its longest dimension; any hole, cut or tear; missing end or pick, thick or thin areas; width less than specified; overall uncleanliness; crease or hard crease; slubs, slugs or gout; shade bar, streaks or uneven dying.
Label: Label missing, incorrect, or illegible. Required information missing or not in accordance with the contract or procurement documents.
Packaging: Not packaged in accordance with the contract or procurement documents.

5.5 Acceptance criteria. Acceptance criteria shall be as specified in the contract or procurement document (see 7.7).

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or purchase order (see 7.7).

7. NOTES.

7.1 Sources of Government documents.

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

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7.1.2 Title 16 CFR Part 303, Rules and Regulations under the Textile Fiber Products Identification Act are available online at https://www.ecfr.gov.

7.1.3 EPA and EO 13693 information is available online at https://www.epa.gov.

7.1.4 Federal Acquisition Regulations (FAR) are available online at https://acquisition.gov.

7.1.5 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1).

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100).
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250).

7.2 Sources for non-Government documents.

7.2.1 AATCC test methods are available for purchase online at https://aatcc.org.

7.2.2 ASQ/ANSI Z1.4 – Sampling Procedures are available online at https://asq.org.

7.2.3 ASTM Standards are available online at https://www.astm.org.

7.3 Intended use. This cloth is intended to be used as pocketing, waistband and lining for dress and service clothing items.

7.4 Standard samples. For access to standard shade samples of the finished cloth, address the contracting activity issuing the invitation for bids or request for proposal.

7.4.1 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 degrees (± 5 degrees) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens."

7.4.2 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

7.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

7.6 Intended use. This cloth is intended to be used as pocketing, waistband and lining for dress and service clothing items.

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7.7 Ordering data. The contract or purchase order should specify the following:

a. Title, number, and date of this CID
b. Class of cloth required (see 2.1)
c. Width of cloth required (see 3.1)
d. Color of cloth required (see 3.1.1)
e. Length put-up required (see 3.6)
f. Fiber identification and labeling information (see 3.7)
g. When toxicity testing is required (see 3.8)
h. Product conformance provisions (see 5.)
i. Acceptance criteria (see 5.5)
j. Packaging requirements (see 6.)

7.8 Key words.
Clothing
Lining
Pocketing
Waistband

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - NU
Air Force – 11

Review activities:
Navy - MC

CIVIL COORDINATING AGENCY ACTIVITY
GSA-FAS

Preparing Activity:
DLA-CT

(Project 8305-2024-005)

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